

Jagjeet Singh

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Summary

AI/NLP engineer and security researcher with a track record of shipping production-grade systems under real-world constraints. IEEE-published author (89.2% precision, 4.2/5 usability), NASA Space Apps Challenge winner (1st/300+ teams), and open-source contributor to active security tooling. Experienced in architecting LLM-backed pipelines, offline-first PWAs, and CI/CD workflows — with a demonstrated bias for measurable impact over theoretical elegance.

Education

Galgotias University

B.Tech in Computer Science and Engineering

Greater Noida, India

2022 – 2026

CGPA: 8.3/10 | Relevant Coursework: Data Structures & Algorithms, Machine Learning, Operating Systems, DBMS

Technical Skills

- **Languages:** Python, Java, JavaScript, Go, SQL
- **AI/ML:** LLMs (Transformers, Gemini API), LangChain, NLTK, TF-IDF, Levenshtein Distance, VADER, Scikit-learn, Pandas, NumPy
- **Web/Systems:** React.js, Firebase (Firestore, RTDB), FastAPI, PWA (Service Workers), REST APIs, Docker, Microservices
- **Cloud/DevOps:** GCP (Cloud Run, Firebase Hosting), GitHub Actions (CI/CD), Vercel, PyPI Packaging
- **Databases:** PostgreSQL, MongoDB, Firebase Realtime DB, LocalStorage
- **Security:** NetExec (nxc), Nmap, LDAP/SMB enumeration, Active Directory, Bug Bounty (HackerOne)
- **Tools:** Git, Postman, jsPDF, PapaParse, pytest, Linux/Kali

Selected Projects

Symptom2Care – Offline-First AI Healthcare Platform

JavaScript, Gemini API, Firebase

- **Architected a hybrid NLP pipeline** combining Google Gemini API (online) and Levenshtein distance matching (offline), maintaining **100% system availability** in low-connectivity environments across 80+ symptom mappings.
- **Achieved 88.3% F1-score** (online) and **74.6% F1-score** (offline) across 50 validated test cases, outperforming baseline keyword matching by **+14.7 F1 points**.
- **Engineered a Progressive Web App (PWA)** with service worker caching strategies, attaining a **90+ Lighthouse score** and reducing second-load latency by **78%** (1.8s → 0.4s).
- **Designed a rule-based emergency detection engine** identifying 12 critical cardiac/respiratory patterns with **100% precision and zero false negatives** across 50+ synthetic validation tests.
- **Integrated Firebase Firestore** for real-time session persistence and PDF export via jsPDF, supporting full offline-to-online data sync on reconnection.

Visual Impaired Assistant – Multilingual Accessible AI Tool

Python, NLTK, SpeechRecognition,

PyPI

- **Architected a bilingual speech-to-text pipeline** with automated English/Hindi language detection and translation, enabling accessible voice-driven interaction for visually impaired users.
- **Implemented a 3-stage NLP processing layer** encompassing POS tagging, Named Entity Recognition (NER), and sentiment analysis (VADER), supporting context-aware command parsing across multiple interaction patterns.

- **Published package to PyPI** (`pip install visual-impaired-assistant`) with fully automated GitHub Actions CI/CD pipeline, enabling one-command installation and reducing manual release overhead.
- **Engineered adaptive audio pre-processing** with noise-robust error recovery and graceful degradation across variable acoustic environments, improving command recognition reliability in real-world conditions.

Open Source Contributions

nxc-scan (*pexmee/nxc-scan*)

Python, NetExec, Active Directory

May 2025

- **Authored and merged PR #4** to `pexmee/nxc-scan`, a CLI automation tool for multi-protocol network scanning (SMB, LDAP, RDP, WinRM) using NetExec — reviewed, live-tested, and **merged by the project maintainer**.
- **Implemented `--batch-ldap` feature** enabling sequential LDAP query batching in a single scan session, reducing manual enumeration steps during Active Directory penetration tests.
- **Validated against live Windows Server 2022 AD environment**, successfully enumerating domain users, GMSA accounts, and SMB signing configuration with confirmed output fidelity.

Research Publications

Symptom2Care: A Big Data-Driven NLP Platform for Preventive Healthcare

IEEE IC3ECSBHI 2026 | DOI: [10.1109/IC3ECSBHI67834.2026.11468987](https://doi.org/10.1109/IC3ECSBHI67834.2026.11468987) | February 2026

Authors: Pawan Kumar, Jagjeet Singh, Smita Parida, Anil Sharma

- **Authored and published** peer-reviewed research in IEEE IC3ECSBHI 2026 validating a hybrid on-line/offline NLP architecture for preventive healthcare in resource-constrained, low-connectivity settings.
- Documented system benchmarks: **89.2% Precision, 87.5% Recall, 88.3% F1-score** (online), and **<1s offline latency** across 50 validated test cases.
- Conducted user acceptance testing with **30 beta users**, achieving a **4.2/5 usability rating**, confirming production-deployment readiness in clinical settings.

Achievements

- **Winner, NASA Space Apps Challenge 2024:** Built a geoscience data visualization platform for Apollo lunar seismic datasets, placing **1st among 300+ competing teams** globally across the challenge.
- **Semi-Finalist, Smart India Hackathon:** Built *Campus Connect*, a resource management system for university infrastructure planning and allocation.
- **Wipro TalentNext Java Full Stack Certification** (Oct 2025) – Credential ID: TNext_SE_25_J_250540253; enterprise Java, Spring Boot, and full-stack web development curriculum.
- **Solved 200+ LeetCode Problems:** Consistent problem-solving across Medium/Hard DSA topics including trees, graphs, and dynamic programming.